JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

IIT JAM 2022 Official Paper

Category: CBT Era (2015–2026) • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
60	100 M	180 Min	MCQ • MSQ • NAT (60 Ques

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	30	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	20	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2022 • Q1 • ID: JAM_2022_Q1) MCQ +1 M Neg: −0.33

Q.1	Consider the 2×2 matrix $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$. If the eighth power of M satisfies $M^8 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$, then the value of x is
(A)	21
(B)	22
(C)	34
(D)	35

Question 2 (JAM 2022 • Q2 • ID: JAM_2022_Q2) MCQ +1 M Neg: −0.33

Q.2	The rank of the 4×6 matrix $\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$ with entries in $\mathbb{R}$, is
(A)	1
(B)	2
(C)	3
(D)	4

Question 3 (JAM 2022 • Q3 • ID: JAM_2022_Q3) MCQ +1 M Neg: −0.33

Q.3	Let V be the real vector space consisting of all polynomials in one variable with real coefficients and having degree at most 6, together with the zero polynomial. Then which one of the following is true?
(A)	$\{f \in V : f(1/2) \notin \mathbb{Q}\}$ is a subspace of V.
(B)	$\{f \in V : f(1/2) = 1\}$ is a subspace of V.
(C)	$\{f \in V : f(1/2) = f(1)\}$ is a subspace of V.
(D)	$\{f \in V : f'(1/2) = 1\}$ is a subspace of V.

Question 4 (JAM 2022 • Q4 • ID: JAM_2022_Q4) MCQ +1 M Neg: −0.33

Q.4	Let G be a group of order 2022. Let H and K be subgroups of G of order 337 and 674, respectively. If $H \cup K$ is also a subgroup of G, then which one of the following is FALSE?
(A)	H is a normal subgroup of $H \cup K$.
(B)	The order of $H \cup K$ is 1011.
(C)	The order of $H \cup K$ is 674.
(D)	K is a normal subgroup of $H \cup K$.

Question 5 (JAM 2022 • Q5 • ID: JAM_2022_Q5) MCQ +1 M Neg: −0.33

Q.5	The radius of convergence of the power series $$\sum_{n=1}^{\infty}\left(\frac{n^3}{4^n}\right)x^{5n}$$ is
(A)	4
(B)	$\sqrt[5]{4}$
(C)	$\frac{1}{4}$
(D)	$\frac{1}{\sqrt[5]{4}}$

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Question 6 (JAM 2022 • Q6 • ID: JAM_2022_Q6) MCQ +1 M Neg: −0.33

Q.6	Let (x_n) and (y_n) be sequences of real numbers defined by $x_1 = 1,\ \ y_1 = \frac{1}{2},\ \ x_{n+1} = \frac{x_n + y_n}{2},\quad$ and $\quad y_{n+1} = \sqrt{x_n y_n}\quad$ for all $n \in \mathbb{N}$. Then which one of the following is true?
(A)	(x_n) is convergent, but (y_n) is not convergent.
(B)	(x_n) is not convergent, but (y_n) is convergent.
(C)	Both (x_n) and (y_n) are convergent and $\lim_{n\to\infty} x_n > \lim_{n\to\infty} y_n$.
(D)	Both (x_n) and (y_n) are convergent and $\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n$.

Question 7 (JAM 2022 • Q7 • ID: JAM_2022_Q7) MCQ +1 M Neg: −0.33

Q.7	Suppose $$a_n = \frac{3^n + 3}{5^n - 5} \quad \text{and} \quad b_n = \frac{1}{(1+n^2)^{\frac{1}{4}}} \quad \text{for } n = 2,3,4,\ldots.$$ Then which one of the following is true?
(A)	Both $\sum_{n=2}^{\infty} a_n$ and $\sum_{n=2}^{\infty} b_n$ are convergent.
(B)	Both $\sum_{n=2}^{\infty} a_n$ and $\sum_{n=2}^{\infty} b_n$ are divergent.
(C)	$\sum_{n=2}^{\infty} a_n$ is convergent and $\sum_{n=2}^{\infty} b_n$ is divergent.
(D)	$\sum_{n=2}^{\infty} a_n$ is divergent and $\sum_{n=2}^{\infty} b_n$ is convergent.

Question 8 (JAM 2022 • Q8 • ID: JAM_2022_Q8) MCQ +1 M Neg: −0.33

Q.8	Consider the series $$\sum_{n=1}^{\infty} \frac{1}{n^m\left(1+\frac{1}{n^p}\right)}$$ where m and p are real numbers. Under which of the following conditions does the above series converge?
(A)	$m > 1$.
(B)	$0 < m < 1$ and $p > 1$.
(C)	$0 < m \leq 1$ and $0 \leq p \leq 1$.
(D)	$m = 1$ and $p > 1$.

Question 9 (JAM 2022 • Q9 • ID: JAM_2022_Q9) MCQ +1 M Neg: −0.33

Q.9	Let c be a positive real number and let $u: \mathbb{R}^2 \to \mathbb{R}$ be defined by $$u(x,t) = \frac{1}{2c}\int_{x-ct}^{x+ct} e^{s^2} ds \quad \text{for} \quad (x,t) \in \mathbb{R}^2.$$ Then which one of the following is true?
(A)	$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ on $\mathbb{R}^2$.
(B)	$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ on $\mathbb{R}^2$.
(C)	$\frac{\partial u}{\partial t}\frac{\partial u}{\partial x} = 0$ on $\mathbb{R}^2$.
(D)	$\frac{\partial^2 u}{\partial t\, \partial x} = 0$ on $\mathbb{R}^2$.

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Question 10 (JAM 2022 • Q10 • ID: JAM_2022_Q10) MCQ +1 M Neg: −0.33

Q.10	Let $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Consider the functions $$u : \mathbb{R}^2 - \{(0,0)\} \to \mathbb{R} \quad \text{and} \quad v : \mathbb{R}^2 - \{(0,0)\} \to \mathbb{R}$$ given by $$u(x,y) = x - \frac{x}{x^2+y^2} \quad \text{and} \quad v(x,y) = y + \frac{y}{x^2+y^2}.$$ The value of the determinant $\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$ at the point $(\cos\theta, \sin\theta)$ is equal to
(A)	$4\sin\theta$.
(B)	$4\cos\theta$.
(C)	$4\sin^2\theta$.
(D)	$4\cos^2\theta$.

Question 11 (JAM 2022 • Q11 • ID: JAM_2022_Q11) MCQ +2 M Neg: −0.67

Q.11	Consider the open rectangle $G = \{(s,t) \in \mathbb{R}^2 : \ 0 < s < 1 \text{ and } 0 < t < 1\}$ and the map $T: G \to \mathbb{R}^2$ given by $$T(s,t) = \left(\frac{\pi s(1-t)}{2}, \ \frac{\pi(1-s)}{2}\right) \quad \text{for } (s,t) \in G\,.$$ Then the area of the image $T(G)$ of the map T is equal to
(A)	$\frac{\pi}{4}$
(B)	$\frac{\pi^2}{4}$
(C)	$\frac{\pi^2}{8}$
(D)	1

Question 12 (JAM 2022 • Q12 • ID: JAM_2022_Q12) MCQ +2 M Neg: −0.67

Q.12	Let T denote the sum of the convergent series $$1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}+\cdots+\frac{(-1)^{n+1}}{n}+\cdots$$ and let S denote the sum of the convergent series $$1-\frac{1}{2}-\frac{1}{4}+\frac{1}{3}-\frac{1}{6}-\frac{1}{8}+\frac{1}{5}-\frac{1}{10}-\frac{1}{12}+\cdots=\sum_{n=1}^{\infty}a_n,$$ where $$a_{3m-2}=\frac{1}{2m-1},\quad a_{3m-1}=\frac{-1}{4m-2}\quad\text{and}\quad a_{3m}=\frac{-1}{4m}\quad\text{for } m\in\mathbb{N}.$$ Then which one of the following is true?
(A)	$T=S$ and $S\neq 0$.
(B)	$2T=S$ and $S\neq 0$.
(C)	$T=2S$ and $S\neq 0$.
(D)	$T=S=0$.

Question 13 (JAM 2022 • Q13 • ID: JAM_2022_Q13) MCQ +2 M Neg: −0.67

Q.13	Let $u: \mathbb{R} \to \mathbb{R}$ be a twice continuously differentiable function such that $u(0) > 0$ and $u'(0) > 0$. Suppose u satisfies $$u''(x) = \frac{u(x)}{1+x^2} \quad \text{for all } x \in \mathbb{R}.$$ Consider the following two statements: I. The function uu' is monotonically increasing on $[0, \infty)$. II. The function u is monotonically increasing on $[0, \infty)$. Then which one of the following is correct?
(A)	Both I and II are false.
(B)	Both I and II are true.
(C)	I is false, but II is true.
(D)	I is true, but II is false.

Question 14 (JAM 2022 • Q14 • ID: JAM_2022_Q14) MCQ +2 M Neg: −0.67

Q.14	The value of $$\lim_{n\to\infty}\sum_{k=2}^{n}\frac{\sqrt{n+1}-\sqrt{n}}{k(\ln k)^2}$$ is equal to
(A)	∞
(B)	1
(C)	e
(D)	0

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Question 15 (JAM 2022 • Q15 • ID: JAM_2022_Q15) MCQ +2 M Neg: −0.67

Q.15	For $t \in \mathbb{R}$, let $[t]$ denote the greatest integer less than or equal to t. Define functions $h: \mathbb{R}^2 \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ by $$h(x,y) = \begin{cases} \frac{-1}{x^2-y} & \text{if } x^2 \neq y, \\ 0 & \text{if } x^2 = y \end{cases} \quad \text{and} \quad g(x) = \begin{cases} \frac{\sin x}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$ Then which one of the following is FALSE?
(A)	$\lim_{(x,y)\to(\sqrt{2},\pi)} \cos\left(\frac{x^2y}{x^2+1}\right) = \frac{-1}{2}.$
(B)	$\lim_{(x,y)\to(\sqrt{2},2)} e^{h(x,y)} = 0.$
(C)	$\lim_{(x,y)\to(e,e)} \ln\left(x^{y-[y]}\right) = e - 2.$
(D)	$\lim_{(x,y)\to(0,0)} e^{2y} g(x) = 1.$

Question 16 (JAM 2022 • Q16 • ID: JAM_2022_Q16) MCQ +2 M Neg: −0.67

Q.16	Let $P \in M_4(\mathbb{R})$ be such that P^4 is the zero matrix, but P^3 is a nonzero matrix. Then which one of the following is FALSE?
(A)	For every nonzero vector $v \in \mathbb{R}^4$, the subset $\{v, Pv, P^2v, P^3v\}$ of the real vector space $\mathbb{R}^4$ is linearly independent.
(B)	The rank of P^k is $4-k$ for every $k \in \{1,2,3,4\}$.
(C)	0 is an eigenvalue of P.
(D)	If $Q \in M_4(\mathbb{R})$ is such that Q^4 is the zero matrix, but Q^3 is a nonzero matrix, then there exists a nonsingular matrix $S \in M_4(\mathbb{R})$ such that $S^{-1}QS = P$.

Question 17 (JAM 2022 • Q17 • ID: JAM_2022_Q17) MCQ +2 M Neg: −0.67

Q.17	For $X, Y \in M_2(\mathbb{R})$, define $(X, Y) = XY - YX$. Let $\mathbf{0} \in M_2(\mathbb{R})$ denote the zero matrix. Consider the two statements: P : $\big(X,(Y,Z)\big) + \big(Y,(Z,X)\big) + \big(Z,(X,Y)\big) = \mathbf{0}$ for all $X, Y, Z \in M_2(\mathbb{R})$. Q : $\big(X,(Y,Z)\big) = \big((X,Y),Z\big)$ for all $X, Y, Z \in M_2(\mathbb{R})$. Then which one of the following is correct?
(A)	Both P and Q are true.
(B)	P is true, but Q is false.
(C)	P is false, but Q is true.
(D)	Both P and Q are false.

Question 18 (JAM 2022 • Q18 • ID: JAM_2022_Q18) MCQ +2 M Neg: −0.67

Q.18	Consider the system of linear equations $$\begin{aligned} x+y+t &= 4, \\ 2x-4t &= 7, \\ x+y+z &= 5, \\ x-3y-z-10t &= \lambda, \end{aligned}$$ where x, y, z, t are variables and λ is a constant. Then which one of the following is true?
(A)	If $\lambda = 1$, then the system has a unique solution.
(B)	If $\lambda = 2$, then the system has infinitely many solutions.
(C)	If $\lambda = 1$, then the system has infinitely many solutions.
(D)	If $\lambda = 2$, then the system has a unique solution.

Question 19 (JAM 2022 • Q19 • ID: JAM_2022_Q19) MCQ +2 M Neg: −0.67

Q.19	Consider the group $(\mathbb{Q}, +)$ and its subgroup $(\mathbb{Z}, +)$. For the quotient group $\mathbb{Q}/\mathbb{Z}$, which one of the following is FALSE?
(A)	$\mathbb{Q}/\mathbb{Z}$ contains a subgroup isomorphic to $(\mathbb{Z}, +)$.
(B)	There is exactly one group homomorphism from $\mathbb{Q}/\mathbb{Z}$ to $(\mathbb{Q}, +)$.
(C)	For all $n \in \mathbb{N}$, there exists $g \in \mathbb{Q}/\mathbb{Z}$ such that the order of g is n.
(D)	$\mathbb{Q}/\mathbb{Z}$ is not a cyclic group.

Question 20 (JAM 2022 • Q20 • ID: JAM_2022_Q20) MCQ +2 M Neg: −0.67

Q.20	For $P \in M_5(\mathbb{R})$ and $i, j \in \{1,2,\ldots,5\}$, let p_{ij} denote the $(i,j)^{\text{th}}$ entry of P. Let $S = \{P \in M_5(\mathbb{R}) : p_{ij} = p_{rs}$ for $i, j, r, s \in \{1,2,\ldots,5\}$ with $i + r = j + s\}$. Then which one of the following is FALSE?
(A)	S is a subspace of the vector space over $\mathbb{R}$ of all 5×5 symmetric matrices.
(B)	The dimension of S over $\mathbb{R}$ is 5.
(C)	The dimension of S over $\mathbb{R}$ is 11.
(D)	If $P \in S$ and all the entries of P are integers, then 5 divides the sum of all the diagonal entries of P.

Question 21 (JAM 2022 • Q21 • ID: JAM_2022_Q21) MCQ +2 M Neg: −0.67

Q.21	On the open interval $(-c, c)$, where c is a positive real number, $y(x)$ is an infinitely differentiable solution of the differential equation $$\frac{dy}{dx} = y^2 - 1 + \cos x,$$ with the initial condition $y(0) = 0$. Then which one of the following is correct?
(A)	$y(x)$ has a local maximum at the origin.
(B)	$y(x)$ has a local minimum at the origin.
(C)	$y(x)$ is strictly increasing on the open interval $(-\delta, \delta)$ for some positive real number δ.
(D)	$y(x)$ is strictly decreasing on the open interval $(-\delta, \delta)$ for some positive real number δ.

Question 22 (JAM 2022 • Q22 • ID: JAM_2022_Q22) MCQ +2 M Neg: −0.67

Q.22	Let $H : \mathbb{R} \to \mathbb{R}$ be the function given by $H(x) = \frac{1}{2}(e^x + e^{-x})$ for $x \in \mathbb{R}$. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $$f(x) = \int_0^{\pi} H(x \sin\theta) d\theta \quad \text{for } x \in \mathbb{R}.$$ Then which one of the following is true?
(A)	$xf''(x) + f'(x) + xf(x) = 0$ for all $x \in \mathbb{R}$.
(B)	$xf''(x) - f'(x) + xf(x) = 0$ for all $x \in \mathbb{R}$.
(C)	$xf''(x) + f'(x) - xf(x) = 0$ for all $x \in \mathbb{R}$.
(D)	$xf''(x) - f'(x) - xf(x) = 0$ for all $x \in \mathbb{R}$.

Question 23 (JAM 2022 • Q23 • ID: JAM_2022_Q23) MCQ +2 M Neg: −0.67

Q.23	Consider the differential equation $$y'' + ay' + y = \sin x \quad \text{for } x \in \mathbb{R}. \qquad (**)$$ Then which one of the following is true?
(A)	If $a = 0$, then all the solutions of $(**)$ are unbounded over $\mathbb{R}$.
(B)	If $a = 1$, then all the solutions of $(**)$ are unbounded over $(0, \infty)$.
(C)	If $a = 1$, then all the solutions of $(**)$ tend to zero as $x \to \infty$.
(D)	If $a = 2$, then all the solutions of $(**)$ are bounded over $(-\infty, 0)$.

Question 24 (JAM 2022 • Q24 • ID: JAM_2022_Q24) MCQ +2 M Neg: −0.67

Q.24	For $g \in \mathbb{Z}$, let $\overline{g} \in \mathbb{Z}_{37}$ denote the residue class of g modulo 37. Consider the group $U_{37} = \{\overline{g} \in \mathbb{Z}_{37} : 1 \leq g \leq 37 \text{ with } \gcd(g, 37) = 1\}$ with respect to multiplication modulo 37. Then which one of the following is FALSE?
(A)	The set $\{\overline{g} \in U_{37} : \overline{g} = (\overline{g})^{-1}\}$ contains exactly 2 elements.
(B)	The order of the element $\overline{10}$ in U_{37} is 36.
(C)	There is exactly one group homomorphism from U_{37} to $(\mathbb{Z}, +)$.
(D)	There is exactly one group homomorphism from U_{37} to $(\mathbb{Q}, +)$.

Question 25 (JAM 2022 • Q25 • ID: JAM_2022_Q25) MCQ +2 M Neg: −0.67

Q.25	For some real number c with $0 < c < 1$, let $\phi: (1-c, 1+c) \to (0, \infty)$ be a differentiable function such that $\phi(1) = 1$ and $y = \phi(x)$ is a solution of the differential equation $$(x^2 + y^2)dx - 4xy\,dy = 0.$$ Then which one of the following is true?
(A)	$\left(3(\phi(x))^2 + x^2\right)^2 = 4x.$
(B)	$\left(3(\phi(x))^2 - x^2\right)^2 = 4x.$
(C)	$\left(3(\phi(x))^2 + x^2\right)^2 = 4\phi(x).$
(D)	$\left(3(\phi(x))^2 - x^2\right)^2 = 4\phi(x).$

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Question 26 (JAM 2022 • Q26 • ID: JAM_2022_Q26) MCQ +2 M Neg: −0.67

Q.26	For a 4×4 matrix $M \in M_4(\mathbb{C})$, let $\overline{M}$ denote the matrix obtained from M by replacing each entry of M by its complex conjugate. Consider the real vector space $$H = \{M \in M_4(\mathbb{C}) : \quad M^{\top} = \overline{M}\}$$ where $M^{\top}$ denotes the transpose of M. The dimension of H as a vector space over $\mathbb{R}$ is equal to
(A)	6
(B)	16
(C)	15
(D)	12

Question 27 (JAM 2022 • Q27 • ID: JAM_2022_Q27) MCQ +2 M Neg: −0.67

Q.27	Let a, b be positive real numbers such that $a < b$. Given that $$\lim_{N\to\infty} \int_0^N e^{-t^2} dt = \frac{\sqrt{\pi}}{2},$$ the value of $$\lim_{N\to\infty} \int_0^N \frac{1}{t^2}\left(e^{-at^2} - e^{-bt^2}\right) dt$$ is equal to
(A)	$\sqrt{\pi}(\sqrt{a} - \sqrt{b})$.
(B)	$\sqrt{\pi}(\sqrt{a} + \sqrt{b})$.
(C)	$-\sqrt{\pi}(\sqrt{a} + \sqrt{b})$.
(D)	$\sqrt{\pi}(\sqrt{b} - \sqrt{a})$.

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Question 28 (JAM 2022 • Q28 • ID: JAM_2022_Q28) MCQ +2 M Neg: −0.67

Q.28	For $-1 \le x \le 1$, if $f(x)$ is the sum of the convergent power series $$x + \frac{x^2}{2^2} + \frac{x^3}{3^2} + \cdots + \frac{x^n}{n^2} + \cdots$$ then $f\left(\frac{1}{2}\right)$ is equal to
(A)	$\int_0^{\frac{1}{2}} \frac{\ln(1-t)}{t}\, dt.$
(B)	$-\int_0^{\frac{1}{2}} \frac{\ln(1-t)}{t}\, dt.$
(C)	$\int_0^{\frac{1}{2}} \mathrm{t}\ln(1+t)\, dt.$
(D)	$\int_0^{\frac{1}{2}} \mathrm{t}\ln(1-t)\, dt.$

Question 29 (JAM 2022 • Q29 • ID: JAM_2022_Q29) MCQ +2 M Neg: −0.67

Q.29	For $n \in \mathbb{N}$ and $x \in [1, \infty)$, let $$f_n(x) = \int_0^{\pi} \left(x^2 + (\cos\theta)\sqrt{x^2 - 1}\right)^n d\theta.$$ Then which one of the following is true?
(A)	$f_n(x)$ is not a polynomial in x if n is odd and $n \geq 3$.
(B)	$f_n(x)$ is not a polynomial in x if n is even and $n \geq 4$.
(C)	$f_n(x)$ is a polynomial in x for all $n \in \mathbb{N}$.
(D)	$f_n(x)$ is not a polynomial in x for any $n \geq 3$.

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Question 30 (JAM 2022 • Q30 • ID: JAM_2022_Q30) MCQ +2 M Neg: −0.67

Q.30	Let P be a 3×3 real matrix having eigenvalues $\lambda_1 = 0, \lambda_2 = 1$ and $\lambda_3 = -1$. Further, $v_1 = \begin{pmatrix}1\\0\\0\end{pmatrix}$, $v_2 = \begin{pmatrix}1\\1\\0\end{pmatrix}$ and $v_3 = \begin{pmatrix}1\\0\\1\end{pmatrix}$ are eigenvectors of the matrix P corresponding to the eigenvalues λ_1, λ_2 and λ_3, respectively. Then the entry in the first row and the third column of P is
(A)	0
(B)	1
(C)	-1
(D)	2

Question 31 (JAM 2022 • Q31 • ID: JAM_2022_Q31) MSQ +2 M Neg: Nil (0)

Q.31	Let $(-c, c)$ be the largest open interval in $\mathbb{R}$ (where c is either a positive real number or $c = \infty$) on which the solution $y(x)$ of the differential equation $$\frac{dy}{dx} = x^2 + y^2 + 1 \quad \text{with initial condition} \quad y(0) = 0$$ exists and is unique. Then which of the following is/are true?
(A)	$y(x)$ is an odd function on $(-c, c)$.
(B)	$y(x)$ is an even function on $(-c, c)$.
(C)	$\big(y(x)\big)^2$ has a local minimum at 0.
(D)	$\big(y(x)\big)^2$ has a local maximum at 0.

Question 32 (JAM 2022 • Q32 • ID: JAM_2022_Q32) MSQ +2 M Neg: Nil (0)

Q.32	Let S be the set of all continuous functions $f:[-1,1] \to \mathbb{R}$ satisfying the following three conditions: (i) f is infinitely differentiable on the open interval $(-1,1)$, (ii) the Taylor series $$f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\cdots$$ of f at 0 converges to $f(x)$ for each $x \in (-1,1)$, (iii) $f\left(\frac{1}{n}\right)=0$ for all $n \in \mathbb{N}$. Then which of the following is/are true?
(A)	$f(0)=0$ for every $f \in S$.
(B)	$f'\left(\frac{1}{2}\right)=0$ for every $f \in S$.
(C)	There exists $f \in S$ such that $f'\left(\frac{1}{2}\right) \neq 0$.
(D)	There exists $f \in S$ such that $f(x) \neq 0$ for some $x \in [-1,1]$.

Question 33 (JAM 2022 • Q33 • ID: JAM_2022_Q33) MSQ +2 M Neg: Nil (0)

Q.33

Define $f:[0,1] \to [0,1]$ by

$$f(x) = \begin{cases} 1 & \text{if } x = 0, \\ \frac{1}{n} & \text{if } x = \frac{m}{n} \text{ for some } m, n \in \mathbb{N} \text{ with } m \leq n \text{ and } \gcd(m,n) = 1, \\ 0 & \text{if } x \in [0,1] \text{ is irrational.} \end{cases}$$

and define $g:[0,1] \to [0,1]$ by

$$g(x) = \begin{cases} 0 & \text{if } x = 0, \\ 1 & \text{if } x \in (0,1]. \end{cases}$$

Then which of the following is/are true?

(A) f is Riemann integrable on $[0,1]$.

(B) g is Riemann integrable on $[0,1]$.

(C) The composite function $f \circ g$ is Riemann integrable on $[0,1]$.

(D) The composite function $g \circ f$ is Riemann integrable on $[0,1]$.

Question 34 (JAM 2022 • Q34 • ID: JAM_2022_Q34) MSQ +2 M Neg: Nil (0)

Q.34	Let S be the set of all functions $f: \mathbb{R} \to \mathbb{R}$ satisfying $$\lvert f(x) - f(y)\rvert^2 \leq \lvert x - y\rvert^3 \quad \text{for all } x, y \in \mathbb{R}.$$ Then which of the following is/are true?
(A)	Every function in S is differentiable.
(B)	There exists a function $f \in S$ such that f is differentiable, but f is not twice differentiable.
(C)	There exists a function $f \in S$ such that f is twice differentiable, but f is not thrice differentiable.
(D)	Every function in S is infinitely differentiable.

Question 35 (JAM 2022 • Q35 • ID: JAM_2022_Q35) MSQ +2 M Neg: Nil (0)

Q.35	A real-valued function $y(x)$ defined on $\mathbb{R}$ is said to be **periodic** if there exists a real number $T > 0$ such that $y(x+T) = y(x)$ for all $x \in \mathbb{R}$. Consider the differential equation $$\frac{d^2y}{dx^2} + 4y = \sin(ax), \quad x \in \mathbb{R}, \qquad (*)$$ where $a \in \mathbb{R}$ is a constant. Then which of the following is/are true?
(A)	All solutions of $(*)$ are periodic for every choice of a.
(B)	All solutions of $(*)$ are periodic for every choice of $a \in \mathbb{R} - \{-2, 2\}$.
(C)	All solutions of $(*)$ are periodic for every choice of $a \in \mathbb{Q} - \{-2, 2\}$.
(D)	If $a \in \mathbb{R} - \mathbb{Q}$, then there is a unique periodic solution of $(*)$.

Question 36 (JAM 2022 • Q36 • ID: JAM_2022_Q36) MSQ +2 M Neg: Nil (0)

Q.36	Let M be a positive real number and let $u, v : \mathbb{R}^2 \to \mathbb{R}$ be continuous functions satisfying $\sqrt{u(x,y)^2 + v(x,y)^2} \geq M\sqrt{x^2+y^2}$ for all $(x,y) \in \mathbb{R}^2$. Let $F: \mathbb{R}^2 \to \mathbb{R}^2$ be given by $F(x,y) = \big(u(x,y), v(x,y)\big)$ for $(x,y) \in \mathbb{R}^2$. Then which of the following is/are true?
(A)	F is injective.
(B)	If K is open in $\mathbb{R}^2$, then $F(K)$ is open in $\mathbb{R}^2$.
(C)	If K is closed in $\mathbb{R}^2$, then $F(K)$ is closed in $\mathbb{R}^2$.
(D)	If E is closed and bounded in $\mathbb{R}^2$, then $F^{-1}(E)$ is closed and bounded in $\mathbb{R}^2$.

Question 37 (JAM 2022 • Q37 • ID: JAM_2022_Q37) MSQ +2 M Neg: Nil (0)

Q.37	Let G be a finite group of order at least two and let e denote the identity element of G. Let $\sigma: G \to G$ be a bijective group homomorphism that satisfies the following two conditions: (i) If $\sigma(g) = g$ for some $g \in G$, then $g = e$, (ii) $(\sigma \circ \sigma)(g) = g$ for all $g \in G$. Then which of the following is/are correct?
(A)	For each $g \in G$, there exists $h \in G$ such that $h^{-1}\sigma(h) = g$.
(B)	There exists $x \in G$ such that $x\sigma(x) \neq e$.
(C)	The map σ satisfies $\sigma(x) = x^{-1}$ for every $x \in G$.
(D)	The order of the group G is an odd number.

Question 38 (JAM 2022 • Q38 • ID: JAM_2022_Q38) MSQ +2 M Neg: Nil (0)

Q.38	Let (x_n) be a sequence of real numbers. Consider the set $$P = \{n \in \mathbb{N} : x_n > x_m \text{ for all } m \in \mathbb{N} \text{ with } m > n\}.$$ Then which of the following is/are true?
(A)	If P is finite, then (x_n) has a monotonically increasing subsequence.
(B)	If P is finite, then no subsequence of (x_n) is monotonically increasing.
(C)	If P is infinite, then (x_n) has a monotonically decreasing subsequence.
(D)	If P is infinite, then no subsequence of (x_n) is monotonically decreasing.

Question 39 (JAM 2022 • Q39 • ID: JAM_2022_Q39) MSQ +2 M Neg: Nil (0)

Q.39	Let V be the real vector space consisting of all polynomials in one variable with real coefficients and having degree at most 5, together with the zero polynomial. Let $T: V \to \mathbb{R}$ be the linear map defined by $T(1) = 1$ and $$T\big(x(x-1)\cdots(x-k+1)\big) = 1 \quad \text{for } 1 \le k \le 5.$$ Then which of the following is/are true?
(A)	$T(x^4) = 15$.
(B)	$T(x^3) = 5$.
(C)	$T(x^4) = 14$.
(D)	$T(x^3) = 3$.

Question 40 (JAM 2022 • Q40 • ID: JAM_2022_Q40) MSQ +2 M Neg: Nil (0)

Q.40	Let P be a fixed 3×3 matrix with entries in $\mathbb{R}$. Which of the following maps from $M_3(\mathbb{R})$ to $M_3(\mathbb{R})$ is/are linear?
(A)	$T_1: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_1(M) = MP - PM$ for $M \in M_3(\mathbb{R})$.
(B)	$T_2: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_2(M) = M^2P - P^2M$ for $M \in M_3(\mathbb{R})$.
(C)	$T_3: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_3(M) = MP^2 + P^2M$ for $M \in M_3(\mathbb{R})$.
(D)	$T_4: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_4(M) = MP^2 - PM^2$ for $M \in M_3(\mathbb{R})$.
Section C: Q.41 – Q.50 Carry ONE mark each.	

Question 41 (JAM 2022 • Q41 • ID: JAM_2022_Q41) NAT +1 M Neg: Nil (0)

Q.41	The value of the limit $$\lim_{n\to\infty}\left(\frac{(1^4+2^4+\cdots+n^4)}{n^5}+\frac{1}{\sqrt{n}}\left(\frac{1}{\sqrt{n+1}}+\frac{1}{\sqrt{n+2}}+\cdots+\frac{1}{\sqrt{4n}}\right)\right)$$ is equal to _____. (*Rounded off to two decimal places*)

Question 42 (JAM 2022 • Q42 • ID: JAM_2022_Q42) NAT +1 M Neg: Nil (0)

Q.42 Consider the function $u: \mathbb{R}^3 \to \mathbb{R}$ given by

$$u(x_1, x_2, x_3) = x_1 x_2^4 x_3^2 - x_1^3 x_3^4 - 26 x_1^2 x_2^2 x_3^3.$$

Let $c \in \mathbb{R}$ and $k \in \mathbb{N}$ be such that

$$x_1 \frac{\partial u}{\partial x_2} + 2x_2 \frac{\partial u}{\partial x_3}$$

evaluated at the point (t, t^2, t^3), equals ct^k for every $t \in \mathbb{R}$. Then the value of k is equal to _____.

Question 43 (JAM 2022 • Q43 • ID: JAM_2022_Q43) NAT +1 M Neg: Nil (0)

Q.43 Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + 3x^2 y = x^2, \quad \text{for } x \in \mathbb{R},$$

satisfying the initial condition $y(0) = 4$.

Then $\lim_{x \to \infty} y(x)$ is equal to _____. (*Rounded off to two decimal places*)

Question 44 (JAM 2022 • Q44 • ID: JAM_2022_Q44) NAT +1 M Neg: Nil (0)

Q.44 The sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{(4n-3)(4n+1)}$$

is equal to _____. (*Rounded off to two decimal places*)

Question 45 (JAM 2022 • Q45 • ID: JAM_2022_Q45) NAT +1 M Neg: Nil (0)

Q.45	The number of distinct subgroups of $\mathbb{Z}_{999}$ is _____.

Question 46 (JAM 2022 • Q46 • ID: JAM_2022_Q46) NAT +1 M Neg: Nil (0)

Q.46	The number of elements of order 12 in the symmetric group S_7 is equal to _____.

Question 47 (JAM 2022 • Q47 • ID: JAM_2022_Q47) NAT +1 M Neg: Nil (0)

Q.47	Let $y(x)$ be the solution of the differential equation $$xy^2y' + y^3 = \frac{\sin x}{x} \quad \text{for } x > 0,$$ satisfying $y\left(\frac{\pi}{2}\right) = 0$. Then the value of $y\left(\frac{5\pi}{2}\right)$ is equal to _____. (*Rounded off to two decimal places*)

Question 48 (JAM 2022 • Q48 • ID: JAM_2022_Q48) NAT +1 M Neg: Nil (0)

Q.48	Consider the region $$G = \{(x, y, z) \in \mathbb{R}^3 : 0 < z < x^2 - y^2,\ x^2 + y^2 < 1\}.$$ Then the volume of G is equal to _____. (*Rounded off to two decimal places*)

Question 49 (JAM 2022 • Q49 • ID: JAM_2022_Q49) NAT +1 M Neg: Nil (0)

Q.49 Given that $y(x)$ is a solution of the differential equation

$$x^2y'' + xy' - 4y = x^2$$

on the interval $(0, \infty)$ such that $\lim_{x \to 0^+} y(x)$ exists and $y(1) = 1$. The value of $y'(1)$ is equal to _____. (*Rounded off to two decimal places*)

Question 50 (JAM 2022 • Q50 • ID: JAM_2022_Q50) NAT +1 M Neg: Nil (0)

Q.50 Consider the family $\mathcal{F}_1$ of curves lying in the region

$$\{(x, y) \in \mathbb{R}^2 : y > 0 \text{ and } 0 < x < \pi\}$$

and given by

$$y = \frac{c(1 - \cos x)}{\sin x}, \quad \text{where } c \text{ is a positive real number.}$$

Let $\mathcal{F}_2$ be the family of orthogonal trajectories to $\mathcal{F}_1$. Consider the curve $\mathcal{C}$ belonging to the family $\mathcal{F}_2$ passing through the point $\left(\frac{\pi}{3}, 1\right)$. If a is a real number such that $\left(\frac{\pi}{4}, a\right)$ lies on $\mathcal{C}$, then the value of a^4 is equal to _____. (*Rounded off to two decimal places*)

Question 51 (JAM 2022 • Q51 • ID: JAM_2022_Q51) NAT +2 M Neg: Nil (0)

Q.51

For $t \in \mathbb{R}$, let $[t]$ denote the greatest integer less than or equal to t.

Let $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 4\}$. Let $f: D \to \mathbb{R}$ and $g: D \to \mathbb{R}$ be defined by $f(0,0) = g(0,0) = 0$ and

$$f(x,y) = [x^2 + y^2]\frac{x^2y^2}{x^4 + y^4}, \qquad g(x,y) = [y^2]\frac{xy}{x^2 + y^2}$$

for $(x, y) \neq (0, 0)$. Let E be the set of points of D at which both f and g are discontinuous. The number of elements in the set E is _____.

Question 52 (JAM 2022 • Q52 • ID: JAM_2022_Q52) NAT +2 M Neg: Nil (0)

Q.52

If G is the region in $\mathbb{R}^2$ given by

$$G = \left\{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1, \ \frac{x}{\sqrt{3}} < y < \sqrt{3}x, \ x > 0, \ y > 0\right\}$$

then the value of

$$\frac{200}{\pi}\iint_G x^2 dx\, dy$$

is equal to _____. (*Rounded off to two decimal places*)

Question 53 (JAM 2022 • Q53 • ID: JAM_2022_Q53) NAT +2 M Neg: Nil (0)

Q.53

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}$ and let $A^{\top}$ denote the transpose of A. Let $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ be column vectors with entries in $\mathbb{R}$ such that $u_1^2 + u_2^2 = 1$ and $v_1^2 + v_2^2 + v_3^2 = 1$. Suppose

$$Au = \sqrt{2}\, v \quad \text{and} \quad A^{\top} v = \sqrt{2}\, u.$$

Then $|u_1 + 2\sqrt{2}\, v_1|$ is equal to _____. (*Rounded off to two decimal places*)

Question 54 (JAM 2022 • Q54 • ID: JAM_2022_Q54) NAT +2 M Neg: Nil (0)

Q.54

Let $f : [0, \pi] \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} (x - \pi)e^{\sin x} & \text{if } 0 \le x \le \dfrac{\pi}{2}, \\ xe^{\sin x} + \dfrac{4}{\pi} & \text{if } \dfrac{\pi}{2} < x \le \pi. \end{cases}$$

Then the value of

$$\int_0^{\pi} f(x)dx$$

is equal to _____. (*Rounded off to two decimal places*)

Question 55 (JAM 2022 • Q55 • ID: JAM_2022_Q55) NAT +2 M Neg: Nil (0)

Q.55 Let r be the radius of convergence of the power series

$$\frac{1}{3}+\frac{x}{5}+\frac{x^2}{3^2}+\frac{x^3}{5^2}+\frac{x^4}{3^3}+\frac{x^5}{5^3}+\frac{x^6}{3^4}+\frac{x^7}{5^4}+\cdots.$$

Then the value of r^2 is equal to _____. (*Rounded off to two decimal places*)

Question 56 (JAM 2022 • Q56 • ID: JAM_2022_Q56) NAT +2 M Neg: Nil (0)

Q.56 Define $f:\mathbb{R}^2\to\mathbb{R}$ by

$$f(x,y)=x^2+2y^2-x \quad \text{for } (x,y)\in\mathbb{R}^2.$$

Let $D=\{(x,y)\in\mathbb{R}^2 : x^2+y^2\le 1\}$ and $E=\left\{(x,y)\in\mathbb{R}^2 : \frac{x^2}{4}+\frac{y^2}{9}\le 1\right\}$.

Consider the sets

$$D_{\max}=\{(a,b)\in D : f \text{ has absolute maximum on } D \text{ at } (a,b)\},$$

$$D_{\min}=\{(a,b)\in D : f \text{ has absolute minimum on } D \text{ at } (a,b)\},$$

$$E_{\max}=\{(c,d)\in E : f \text{ has absolute maximum on } E \text{ at } (c,d)\},$$

$$E_{\min}=\{(c,d)\in E : f \text{ has absolute minimum on } E \text{ at } (c,d)\}.$$

Then the total number of elements in the set

$$D_{\max}\cup D_{\min}\cup E_{\max}\cup E_{\min}$$

is equal to _____.

Question 57 (JAM 2022 • Q57 • ID: JAM_2022_Q57) NAT +2 M Neg: Nil (0)

Q.57

Consider the 4×4 matrix $M = \begin{pmatrix} 11 & 10 & 10 & 10 \\ 10 & 11 & 10 & 10 \\ 10 & 10 & 11 & 10 \\ 10 & 10 & 10 & 11 \end{pmatrix}$. Then the value of the determinant of M is equal to _____.

Question 58 (JAM 2022 • Q58 • ID: JAM_2022_Q58) NAT +2 M Neg: Nil (0)

Q.58

Let σ be the permutation in the symmetric group S_5 given by

$$\sigma(1) = 2, \quad \sigma(2) = 3, \quad \sigma(3) = 1, \quad \sigma(4) = 5, \quad \sigma(5) = 4.$$

Define

$$N(\sigma) = \{\tau \in S_5 : \sigma \circ \tau = \tau \circ \sigma\}.$$

Then the number of elements in $N(\sigma)$ is equal to _____.

Question 59 (JAM 2022 • Q59 • ID: JAM_2022_Q59) NAT +2 M Neg: Nil (0)

Q.59

Let $f:(-1,1) \to \mathbb{R}$ and $g:(-1,1) \to \mathbb{R}$ be thrice continuously differentiable functions such that $f(x) \neq g(x)$ for every nonzero $x \in (-1,1)$. Suppose

$$f(0) = \ln 2, \quad f'(0) = \pi, \quad f''(0) = \pi^2, \quad \text{and} \quad f'''(0) = \pi^9$$

and

$$g(0) = \ln 2, \quad g'(0) = \pi, \quad g''(0) = \pi^2, \quad \text{and} \quad g'''(0) = \pi^3.$$

Then the value of the limit

$$\lim_{x \to 0} \frac{e^{f(x)} - e^{g(x)}}{f(x) - g(x)}$$

is equal to _____. (*Rounded off to two decimal places*)

Question 60 (JAM 2022 • Q60 • ID: JAM_2022_Q60) NAT +2 M Neg: Nil (0)

Q.60

If $f:[0,\infty)\to\mathbb{R}$ and $g:[0,\infty)\to[0,\infty)$ are continuous functions such that

$$\int_0^{x^3+x^2} f(t)dt = x^2 \text{ and } \int_0^{g(x)} t^2dt = 9(x+1)^3 \text{ for all } x \in [0,\infty),$$

then the value of

$$f(2)+g(2)+16\,f(12)$$

is equal to _____. (*Rounded off to two decimal places*)

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2022 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+1	**C**
2	MCQ	+1	**D**
3	MCQ	+1	**C**
4	MCQ	+1	**B**
5	MCQ	+1	**B**
6	MCQ	+1	**D**
7	MCQ	+1	**C**
8	MCQ	+1	**A**
9	MCQ	+1	**A**
10	MCQ	+1	**D**
11	MCQ	+2	**C**
12	MCQ	+2	**C**
13	MCQ	+2	**B**
14	MCQ	+2	**D**
15	MCQ	+2	**Marks to All**
16	MCQ	+2	**A**
17	MCQ	+2	**B**
18	MCQ	+2	**C**
19	MCQ	+2	**A**
20	MCQ	+2	**C**

Q. No.	Type	Marks	Official Key
21	MCQ	+2	**D**
22	MCQ	+2	**C**
23	MCQ	+2	**A**
24	MCQ	+2	**B**
25	MCQ	+2	**B**
26	MCQ	+2	**B**
27	MCQ	+2	**D**
28	MCQ	+2	**B**
29	MCQ	+2	**C**
30	MCQ	+2	**C**
31	MSQ	+2	**A,C**
32	MSQ	+2	**A,B**
33	MSQ	+2	**A,B,C**
34	MSQ	+2	**A,D**
35	MSQ	+2	**C,D**
36	MSQ	+2	**C,D**
37	MSQ	+2	**A,C,D**
38	MSQ	+2	**A,C**
39	MSQ	+2	**A,B**
40	MSQ	+2	**A,C**

Q. No.	Type	Marks	Official Key
41	NAT	+1	**2.19 to 2.21**
42	NAT	+1	**14 to 14**
43	NAT	+1	**0.32 to 0.34**
44	NAT	+1	**0.24 to 0.26**
45	NAT	+1	**8 to 8**
46	NAT	+1	**420 to 420**
47	NAT	+1	**0 to 0**
48	NAT	+1	**0.49 to 0.51**
49	NAT	+1	**2.24 to 2.26**
50	NAT	+1	**2.00 to 2.00**
51	NAT	+2	**18 to 18**
52	NAT	+2	**4.15 to 4.17**
53	NAT	+2	**3.00 to 3.00**
54	NAT	+2	**2.00 to 2.00**
55	NAT	+2	**3.00 to 3.00**
56	NAT	+2	**5 to 5**
57	NAT	+2	**41 to 41**
58	NAT	+2	**6 to 6**
59	NAT	+2	**2.00 to 2.00**
60	NAT	+2	**13.39 to 13.41**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.